

TIVYAKU, Russian Federation

WMO: 317540

Lat: 48.600N

Lon: 137.050E

Elev: 1321

StdP: 14.01

Time Zone: 10.00 (E10)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-35.7	-31.7	-39.3	0.6	-35.2	-35.5	0.8	-31.3	15.5	0.0	13.4	1.3	0.0	230	0.591

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.3	84.1	68.6	80.7	67.2	77.6	65.6	71.6	79.1	69.8	76.9	68.0	74.8	3.8	340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
69.3	113.1	75.1	67.5	106.0	73.4	65.5	99.0	71.6	36.4	79.0	34.8	76.7	33.2	74.7	83.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.6	12.2	11.0	-41.8	89.7	3.1	2.9	-44.0	91.8	-45.8	93.4	-47.6	95.0	-49.9	97.1
			-41.9	76.5	3.1	3.1	-44.1	78.7	-45.9	80.5	-47.6	82.2	-49.8	84.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	30.6	-9.0	-2.5	12.8	33.2	48.5	58.8	66.0	63.2	51.7	34.9	12.4	-5.2	(6)
(7)		DBStd	27.86	9.06	9.76	10.03	8.54	6.87	6.46	4.74	5.15	6.95	7.89	11.61	9.71	(7)
(8)		HDD50	8443	1828	1470	1153	505	112	8	0	1	59	470	1127	1710	(8)
(9)		HDD65	12695	2293	1890	1618	954	511	203	46	95	400	933	1577	2175	(9)
(10)		CDD50	1357	0	0	0	2	66	271	495	411	111	1	0	0	(10)
(11)		CDD65	134	0	0	0	0	0	16	76	40	2	0	0	0	(11)
(12)		CDH74	1563	0	0	0	3	82	381	779	288	29	1	0	0	(12)
(13)		CDH80	342	0	0	0	0	13	93	200	35	1	0	0	0	(13)
(14)	Wind	WSAvg	2.3	1.5	2.1	2.8	2.8	2.4	1.7	1.8	2.1	2.5	2.7	2.0	(14)	
(15)	Precipitation	PrecAvg	32.2	0.7	0.7	1.4	1.8	2.8	3.6	5.7	6.3	3.6	2.1	1.8	1.2	(15)
(16)		PrecMax	45.9	1.4	2.2	5.6	3.5	6.4	6.9	12.9	11.5	9.4	4.7	3.9	2.7	(16)
(17)		PrecMin	22.2	0.3	0.1	0.2	0.4	1.2	0.6	0.6	2.2	1.1	0.5	0.4	0.0	(17)
(18)		PrecStd	6.3	0.4	0.4	1.1	0.8	1.4	1.7	3.1	2.4	1.9	1.1	0.8	0.6	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.0	31.9	47.4	69.5	81.8	87.4	89.1	84.3	77.8	67.3	46.1	29.3	(19)
(20)			MCWB	19.5	27.6	36.3	50.4	60.3	68.4	72.2	70.9	65.1	53.5	39.1	27.1	(20)
(21)		2%	DB	15.3	25.6	40.6	61.2	76.0	82.5	85.2	80.3	73.2	58.4	39.4	21.0	(21)
(22)			MCWB	13.6	21.9	32.8	46.5	56.9	65.8	70.4	68.8	62.0	48.1	35.0	19.3	(22)
(23)		5%	DB	10.8	20.2	35.7	54.1	71.4	78.2	81.9	77.3	69.4	53.0	35.2	15.1	(23)
(24)			MCWB	9.1	17.6	29.8	41.8	55.4	64.2	68.9	67.0	59.6	45.5	32.9	13.8	(24)
(25)		10%	DB	6.6	15.6	31.7	48.8	66.3	74.3	78.5	74.4	65.1	48.7	31.0	10.5	(25)
(26)			MCWB	5.4	13.2	27.3	39.1	52.4	62.2	67.5	66.1	57.7	43.0	28.9	9.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	20.0	27.4	37.2	52.0	62.5	71.9	74.8	73.4	67.0	54.3	40.6	27.2	(27)
(28)			MCD B	22.0	31.7	45.2	67.5	77.0	80.8	83.4	80.6	75.2	63.8	44.5	29.3	(28)
(29)		2%	WB	13.9	21.8	33.7	46.9	59.1	68.3	72.1	71.1	63.9	50.2	35.8	19.6	(29)
(30)			MCD B	15.4	25.5	39.0	58.7	72.2	78.8	81.1	76.5	70.5	56.1	37.8	21.0	(30)
(31)		5%	WB	9.3	17.8	30.5	43.1	56.4	65.9	70.4	69.4	61.3	46.7	32.5	14.0	(31)
(32)			MCD B	10.7	20.2	35.2	52.1	68.2	75.2	78.8	74.5	67.2	51.4	35.1	15.2	(32)
(33)		10%	WB	5.4	13.4	27.4	39.9	53.9	63.6	68.7	67.5	58.7	43.7	28.8	9.3	(33)
(34)			MCD B	6.5	15.6	31.7	47.6	64.0	72.1	76.3	72.6	63.4	48.1	31.1	10.6	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	25.4	30.2	29.3	23.6	26.6	25.7	22.3	19.8	23.5	21.6	19.4	21.2	(35)
(36)			MCDBR	27.9	33.4	32.7	35.1	39.6	34.1	29.5	25.2	30.2	31.4	22.6	22.8	(36)
(37)			MCWBR	26.0	29.3	25.5	21.5	22.1	19.2	15.6	14.0	18.5	20.8	18.5	21.0	(37)
(38)		5% WB	MCD BR	27.3	31.4	28.3	31.4	33.4	28.6	24.3	20.0	24.8	27.5	20.7	21.6	(38)
(39)			MCWBR	25.5	27.8	22.3	20.5	20.2	18.0	14.3	12.8	16.0	19.6	17.3	20.3	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.282	0.330	0.398	0.505	0.539	0.494	0.479	0.429	0.396	0.414	0.365	0.294	(40)	
(41)		taud	2.092	1.951	1.822	1.750	1.810	2.027	2.122	2.273	2.281	2.039	2.020	2.057	(41)	
(42)		Ebn at Noon	239	249	249	235	233	247	249	257	253	216	198	213	(42)	
(43)		Edh at Noon	28	41	55	66	65	53	47	39	35	37	30	25	(43)	
(44)	All-Sky Solar Radiation	RadAvg	579	956	1381	1444	1633	1750	1654	1411	1131	737	471	430	(44)	
(45)		RadStd	24	30	93	144	131	205	142	97	83	63	47	23	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	+0.68	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (6 neighbors)	+0.47	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page